

User Experience Factors and attributes on the Stretchable Display applied Automotive

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Abstract

The stretchable display has design characteristics that it can do multi-axis transformation, express curved surface, and provide 3D interaction and physical tactual experience. Focused on the automotive field, which is one of the major application fields of the stretchable display, the experience factors and detailed attributes that user expects when the stretchable display is applied. Above all, through the literature on the stretchable display interface, the distinctive design attributes are studied, and the user research analysis standard is built by organizing the detailed factors from the preceding automotive user experience literature. On the first user research, the user expectation experience factors and the detailed attributes are analyzed and defined by exploratory collection of data by conducting contextual research and in-depth interview. By adding the expectation experience attributes organized from the preceding literature to the new 37 expectation experience attributes derived from this, 60 experience attributes are organized, and they are analyzed into 13 factors. On the second user research, the positive verification is conducted on these through online survey and confirmatory factor analysis. Finally, 13 factors ('safety', 'high quality', 'multi-sensory', 'pleasure of novelty', 'contextual control convenience', 'physical/mental freedom', 'relaxation service spatiality', 'productive activity spatiality', 'uniqueness', 'aesthetical fashion', 'interface expandability', 'continuous improvement', 'human/object communication') and 54 attributes are organized. Deduced user experience factors and detailed attributes can be provided as a purpose of the stretchable display design on automotive, provided as an evaluation standard after the design, and used as an index for the technical development direction.

Keywords: Stretchable Display, UX Design on Automotive Field, User Experience, User Experience Design

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1 Introduction

As the potential needs for a new form factor based on the user's mobility increase and the smartification of smart devices emerge, the new generation display development is characterized to be Thin, Light, Flexible, and Free Form. Glass-based flat and curved display to plastic-based flexible display types such as Bendable, Foldable, and Rollable displays have been developed, and as next step of flexible display, stretchable technology, which can shift in free form, has been developed[1]. The stretchable display technique was chosen as one of 10 emerging technologies from KISTEP(Korea Institute of S&T Evaluation and Planning) and is getting a focus as a future growth engine business that can pioneer new markets[2]. Depending on the form characteristic that can be installed on various non-planar surfaces based on elasticity, it is forecasted that the use of the stretchable display in the automotive field development potential is high. By overcoming the design restrictions on the curve of the automotive dashboard and foreside and side glass, delivering information on a whole-wall display or rendering a sensible interior design would be effective[3].

Recent automotive display design trends show that the switch from the analog dashboard to the digital cockpit with the LCD or OLED display applied, and this allowed to provide the entertainment features in addition to the road and weather information, resulting in the usage improvement and size increase of the automotive display. As Global market information institute, HIS Markit forecasted that the sales proportion of 10 inches or larger display of the automotive display market to increase from 36.2% in 2019 to 48.2% in 2025[4], the display in the automotive field tends to become larger. However, because the inner surface of the vehicle has amorphous curves, the existing flat display type has confines for utilizing the large area as a display.

There are some cases that a display is applied to where it isn't currently a display of a vehicle, and Samsung Electronics showcased the display installation at the back of the front seat heads, which makes the communication with the passengers at the back seat smoother and supports the passenger's entertainment use[5]. Continental added display to A-Pillar to remove blind spots and increase visual expandability, and there are many cases of replacing the side mirrors with the display[6]. Mercedes-Benz showcased the door used as a

screen on a concept car with self-driving technology[7]. The display form currently offered is – due to various curved surfaces of the vehicle interior – divided into multiple displays and installed; however, as the stretchable display has flexibility and can wrap up complex curved surfaces, it would overcome current display confines and provide new user interfaces and user experiences.

This digital display expansion substitutes physical control button with soft button. Tesla takes OTA(Over The Air) update method to update in-vehicle functions, and it controls most functions of the vehicle with the 15 or 17 inch display[8]. The soft button on the digital display is more effective for applying function update than the fixed physical button, so the use of digital display would increase. However, when the driver operates the touch screen while driving, they might not recognize if the soft button is pressed or faulty operation can happen, so it is questioned that the soft button can be more dangerous than the physical button. It is required to verify expectable user experience on whether the stretchable display free form shape expression characteristic can resolve current display limits based on its design character.

As the field of automotive is undergoing the industry paradigm shifted to be Electronic, Autonomous and Connected, the importance of the HMI (Human Machine Interface) for the vehicle and the driver mutual communication is more stressed, and the stretchable display can be a technology which can secure the competitiveness on service by helping the interaction between the user and the vehicle and providing new user experience. The stretchable display technology is being visualized; in 2017, Samsung Display unveiled 9.1-inch stretchable panel prototype at SID (The Society for Information Display) Display Week[9] and in June 2021, on the academic journal Science Advances, the stretchable display technology, which retains OLED element performance while being able to re-formed freely like rubber, was published[10]. Display targets 2024 to develop product technology of 20% expandable stretchable panel on the curved surfaces for automotive and wearable, with corporate with 21 universities and research institutions[11].

While the research on technical aspects on the stretchable display becomes materialized and make progresses and outcomes, the study on the detailed user experience that can be expected through the advantages of the display from the user side has not been conducted. The user experience is a core factor that determines success or failure in the market, and when the product or service provide experience that the user expects, the consumer receptivity increases and create a new market. Therefore, in this study, it aims to provide the standards and directions for the automotive products with the stretchable display by studying the experience attributes and factors the user expects. To obtain this, the research on literature to understand the distinctive design properties of the stretchable display would be done and the details from the automotive user experience on the preceding literature would be summarized to arrange user research analysis standards. On the first user research, user expectation experience factors and detailed attributes will be analyzed and defined by exploratory

collection of the data through the factors and detailed attributes of the user expectation experience through the contextual research and in-depth interview. On the second user research, the factors and detailed attributes will be empirically verified by researchs and statistical analysis.

2 Related work

2.1 Stretchable Display Design Properties

A stretchable Display is a display that the panel can expand flexibly; the display has elasticity, so when stretched or pressed, the panel expands or is sunken but returns to its original flat panel. This property is rendered with the technical evolution of the Micro LED, substrate and encapsulation layer, TFT driven element, wiring technology, etc. The stretchable display is expected to have the display technology that a pixel is formed with subminiature LED of microunit, and with the Micro LED technology, the LED array structure can be directly laid onto a highly stretchable substrate. It has higher durability than OLED technology, so when panel shape changes, the rate of efficiency deterioration is low, and various materials can be used for the substrate[12]. Since the substrate should be able to stand the high-temperature processing while maintaining flexibility, a method of processing polyimide substrate structure into net shape[13] or utilizing the supplementary substrate such as PDMS (Polydimethylsiloxane) as being pre-expanded, with the backplane and light-emitting layer attached is studied[14]. On the flexible substrate, it is necessary to minimize the pressure on the TFT-driven element, and it is suggested to intensively arrange the elements on a hard area created on the highly flexible substrate. To prevent that the wiring is easily broken when the substrate shape changes, the jolt-proof design of the spring or wave structure is used for the TFT inter-element demonstration signal connection wiring[15]. With these technical elements, the stretchable display has the following distinct form characteristics.

Multi-axis transformation and multi-curved surface form expression. Foldable or rollable display has repetitive single-axis transformation to a specific direction, and stretchable display can have repetitive multi-axis transformation of bending or folding freely in various direction[3]. This characteristic allows the display form transformation depending on the interaction or context change, so one display can express various forms and render tailored interface. Contrary to the fact that the existing display only expressed flat or simple curve shape, it is possible to express complex shape and organic shape and can be applied to various objects interfaces. By applying the real object shape to show virtual images or information, it is possible to implement a realistic interface.

3D interaction and physical tactile experience. Stretchable display allows natural gesture based on the shape transformation such as stretching or twist[16] or kinetic interaction such as push in or pull using the 3-dimensional

transformation[17]. This provides different ways of providing information by providing different action interface based on the depth pushed in or pulled by hands[18] or providing distinct functions with the Z axis depth[19]. When using actuator, by controlling the Z axis height, a kinetic interface can be implemented such as 3-dimensional physical button[20]. The Z axis transformation of the display has the potentials of the designs of button utilization using functional effects and design the texture utilization for sensible effects, and the physical tactile interface can be implemented.

These properties of the stretchable display are expected to provide new interface and ways to interact, and the high degree of freedom on design would provide new user experience by overcoming the limitations of the existing displays.

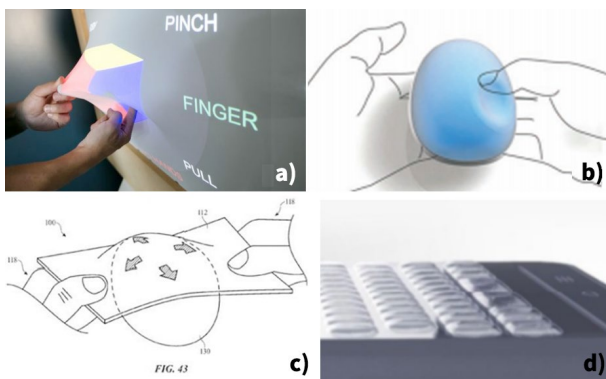


Figure 1 Form characteristics of the stretchable display. (a) The prototype of an elastic, deformable display[18]. (b) Organic User Interface[21]. (c) Stretchable display patent of Apple[22]. (d) Stretchable display that can merge with touch technology[20]

2.2 User experience on automotive

To create a product that user prefers and accepted in the market, the experience which user expects and put values in should be understood. In the existing automotive field, user experience has been considered significant and studied, based on the vehicle segment: compact, mid-size, Executive or Premium, the expected user experience is different[23]. The stretchable display would be applied to the premium cars first due to its official price, and as the consumers owning the premium cars have high preference on the high-class interior design, they are suitable for the new generation display.

In this study, the user experiences from the literature on user behavior and user demands on premium vehicle have been studied. The premium vehicle related user expectation experiences types are 'high quality', 'uniqueness', 'sense/sensibility', 'control', 'self-directed pleasure', 'personalization', 'social prestige'. When the compact and mid-size vehicle segments weigh more functional aspects such as safety and convenience, comfort of seats, etc., the premium vehicle considers individual's emotional satisfaction more important(Table 1).

Table 1 User experience on premium vehicle from literature

Types	User Experience
High quality	high end ^a , quality of technique ^c , superior ^{a, b}
Uniqueness	uniqueness ^{a, c} , customized ^a
Sense/Sensibility	Sensory ^b , emotional ^b
Control	controlling behavior ^b , leading usability ^b
Self-directed pleasure	self-directed symbolic value ^b , hedonic ^{b, c}
Social Prestige	social recognition ^{b, c} , relational ^b , outer-directed symbolic/expressive value ^b , conspicuous ^c

^a Shende, 2014[23]. ^b Tynan, McKechnie, & Hartley, 2014[24]. ^c Sari & Kusuma, 2014[25].

Technology environmental change significantly affects user experience. One of the latest factors that affect user experience changes is the self-driving technology, because it makes the main user behavior of the driver as supplement/non-essential behavior. Society of Automotive Engineers (SAE) J3016 Levels of Driving Automation is divided into 6 level (level 0 to 5) depending on whether the driving (control subject, liability, environment monitoring, emergency response while driving) subject is human or system. For level 0 to 2, liability falls to the driver, and from level 3, system is the main driver and human assists. Level 3 is Conditional Automation, so the driver should be ready to intervene when requested for emergency. Level 4 is High Automation, and the automated driving features drive the vehicle and the driver needs to be on standby in most situations, and level 5 is Full Automation that human driving is eliminated. Level 3 or higher is permitted in 17 States of United States, and Level 3 test driving is permitted on major Europe countries such France, German, and England. In Beijing and Changzhou, China, level 4 test driving is permitted. Republic of Korea permitted the level 3 vehicles since 2020 and is targeting to allow level 4 test driving in 2024[26][27]. Global Autonomous Driving Industry Outlook of Frost & Sullivan from 2019 forecasted that the sales ratio of level 3 self-driving vehicle would be 25% in 2030 and level 4 self-driving technology would lead the automotive industry development[28]. Considering stretchable display prototype development and timing for commercialization, stretchable display would be applied to self-driving vehicles.

In addition, user experience from the literature that discovered user demand and expected user behavior for self-driving technology was studied. Self-driving related user experience types are 'safety', 'independent mobility', 'leisure activity', 'productive activity in transportation', 'physical/mental freedom', 'novelty technology', 'social prestige'. The level of credibility on the automated driving technology is low now, and confidence on safety issue appears critical. As less human driving is involved, space in the care is expected to provide relaxation and productive activities and social activities. Various conveniences and new user experiences over simple means of transportation are required, and various services and user interfaces are expected(Table 2).

Table 2 User experience on self-driving vehicle from literature

Types	User Experience
Safety	safety ^{a,b,c,d} , trust ^{b,e}
Independent mobility	mobility ^c , transport drunk, impaired or under medication ^a , independent mobility for non-drivers ^{a,b} , comparability to public ^b
relaxation	relaxation ^d , comfort ^d
Productive activity in transportation	multitask ^{a,c} , productive while traveling ^b , time management and efficiency ^d
Physical/mental freedom	freedom ^d , transport without having to care ^a , perceived ease of use ^e , emotion and energy management ^d , relieve stress of driving ^b
Novelty technology	new technology ^b , novelty ^d
Social Prestige	being social in the car ^d , being part of the new era ^d

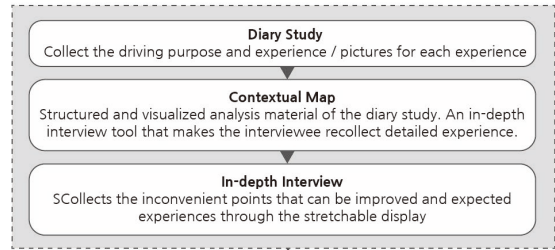
^a König & Neumayr, 2017[29]. ^b Zmud & Sener, 2017[30]. ^c Howard & Dai, 2014[31]. ^d Pettersson, 2017[32]. ^e Rödel, Stadler, Meschtscherjakov, & Tscheligi, 2014[33].

The user experiences are studied from the literature on premium vehicle and self-driving vehicle that stretchable display can be applied. The appearance of new technology affects user experience, similar to self-driving technology affecting user experience, and stretchable display technology would affect user experience as well. This study conducted research and collected user expectation experience to verify what user experiences are expected when stretchable display is applied to automotive field and whether desire and expectation experience that were not found from existing automotive field studies would appear.

3 First User Research

PURPOSE OF FIRST USER SURVEY : Collect the expectation experience attributes on the automotive products having the stretchable display. Categorize the literature-based factors and organize attributes.

Data collection



Data Analysis

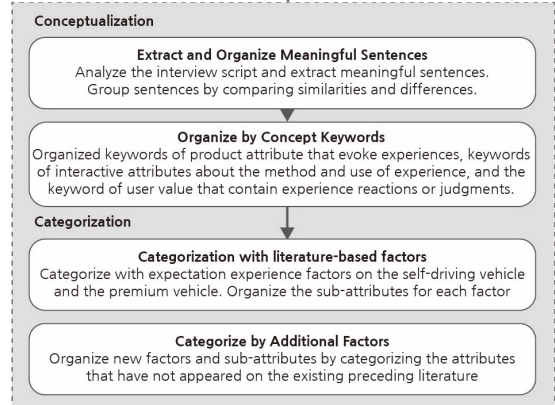


Figure 2 First User Research Purpose and Methods

3.1 Data collection

The purpose of this research is to derive expectation experience attributes on stretchable-display implemented vehicle. In-dept interview along with diary study was conducted. Research targets were six men and women,



Figure 3 First user research process: (a) Context data collection through diary study (b) Analysis with contextual map (c) Collect expected experience with in-depth interview

between twenties and forties, who are interested in future technology, drive partial self-driving vehicle, and join automotive community activity for interest in vehicles.

With the difficulty describing the expectation on future, the expectation experience was collected based on specific context. For preceding research, diary study was conducted. Its format was made to record driving log (based on driving purpose, driving experience is recorded and related pictures are taken and attached) and the issue of used equipment and digital devices. By analyzing data collected from the diary study and subdividing into Task, Activity, Information, Experience, Point, six contextual maps are created with details of driving phase and passenger status.

The semi-structured in-depth interview was conducted with sharing the expectations freely on stretchable display implemented automotive field. The stretchable display is at its beginning stage, so Long Range Forecast technique was used. This technique shows some video clips and prototypes to participants and obtains information from their free talking on the inspirations drawn[34]. The participant indirectly experienced the stretchable display by touching Low fidelity prototype and viewing visual materials, and they shared the inconvenient points that can be improved and expected experiences from the contextual map created previously.

3.2 User experience attributes and factors derivation

To analyze the data, Open coding, which is used to draw thesis from qualitative research, is used. Open coding reveals the meaning of thoughts and ideas in quantitative research and by subdividing data to compare similarity and differences and conceptualize the meaningful contents. Also, notionally similar or semantically related information is gathered to a more abstract concept, Categories[35].

In this study, primary user research interview script data is factorized and detailed contents are analyzed, and contents with meaningful user experience attributes are organized into key words. The specific attributes of user experience represent aspects of the product that causes experience, aspects of interaction about the methods and activities for experience, and aspects of humans including the reaction or judgement to the experience[36]. The meaningful sentences containing these attributes are extracted, and the sentences with similar meaning are organized into keywords. After the analysis, 37 expectation experience attributes are derived for the stretchable display implemented automotive environment.

These attributes are categorized into factors by their similarity or relations. The attributes that can be categorized into the experience factors appeared in the existing preceding literature are combined with the ones in the existing literature, and other factors are organized as new factors. First, the experience factors from the preceding user experience research are categorized into 'safety(F1)', 'high quality(F2)', 'sense/sensibility(F3)', 'novelty(F4)', 'control convenience (F5)', 'physical/mental freedom(F6)', 'relaxation (F7)', 'productive activity in transportation(F8)', 'uniqueness(F9)'. Next, new factors that could not be categorized into the existing preceding literature are 'aesthetic fashion(F10)',

'interface expandability(F11)', 'continuous improvement (F11)', 'human/object communication(F12)'. As a result, there are 13 factors and 60 sub attributes are created(Table 3).

Table 3 Researched user experience factors and attributes

Factor	A.no	Attributes
F1	F1_1	Control Safety ^a
	F1_2	Durability ^a
	F1_3	Eye breakaway prevention ^a
	F1_4	Emergency preparation ^a
	F1_5	Anxiety relief ^a
	F1_6	trust
F2	F2_1	Luxurious material ^a
	F2_2	high end
	F2_3	quality of technique y
	F2_4	superior
F3	F3_1	Five sense stimulus experience ^a
	F3_2	Interaction using five senses ^a
	F3_3	Sensory
	F3_4	emotional
F4	F4_1	Pleasure of new way of control ^a
	F4_2	new technology
	F4_3	novelty
F5	F5_1	Function integration within operational range ^a
	F5_2	Simple control process ^a
	F5_3	Tactual control utilization ^a
	F5_4	Intuitive control ^a
	F5_5	Circumstance customization ^a
	F5_6	Environment development ^a
	F5_7	controlling behavior
	F5_8	leading usability
F6	F6_1	Relieve stress of driving ^a
	F6_2	Admire scenery ^a
	F6_3	Freedom
	F6_4	transport without having to care
	F6_5	perceived ease of use
	F6_6	emotion and energy management
F7	F7_1	Relaxation/convenience space support ^a
	F7_2	Relaxation/convenience service support ^a
	F7_3	Spatiality improvement for storage ^a
	F7_4	Provide independent space ^a
	F7_5	Entertainment ^a
	F7_6	rest while traveling
	F7_7	relaxation
	F7_8	comfort
F8	F8_1	Business/meeting space support ^a
	F8_2	Amenities support ^a
	F8_3	Multitask
	F8_4	productive while traveling
	F8_5	time management and efficiency
F9	F9_1	Self-expression ^a
	F9_2	uniqueness
	F9_3	customized
	F9_4	suitability
F10	F10_1	Color and pattern variation ^a

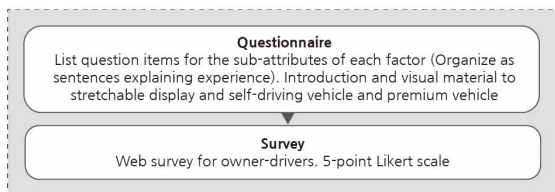
	F10_2	Neat interior design ^a
F11	F11_1	Multi-interface ^a
	F11_2	Digitalization of physical space ^a
	F11_3	Display area expansion ^a
F12	F12_1	Software update ^a
	F12_2	Software utilization ^a
F13	F13_1	External communication ^a
	F13_2	Internal communication ^a
	F13_3	Passenger status recognition ^a
	F13_4	Information shareability ^a
	F13_5	Device paring ^a

^a New user experience attributes & ^b New user experience factor derived from user research

4 Second User Research

PPURPOSE OF SECONDARY SURVEY : Verify the results of the primary survey. Derive final experience factors and sub-attributes through the survey and the confirmatory factor analysis.

Data collection



Data Analysis

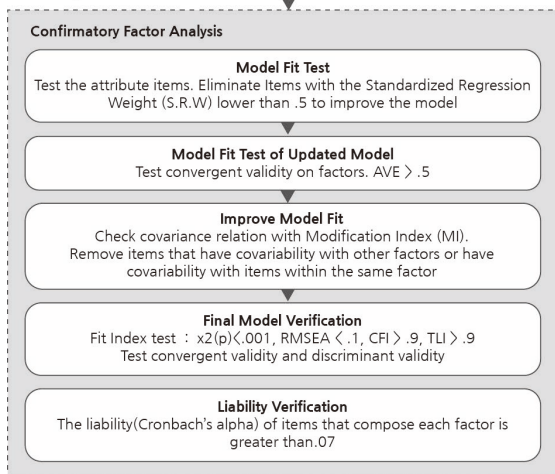


Figure 4 Second User Research Purpose and Methods

4.1 Data collection

Second research was connected to empirically test 13 user experience factors and sub attributes derived from the first research. An online survey was selected for this purpose, and it was conducted through a web survey system in March 2021. Participants are selected with the questions 1) if they include vehicle when selecting possessed devices, and 2) whether they drive themselves. 250 out of 336 are selected to participate on the survey (Gender – Male : 171(68.4%), Female : 79(31.6%). Age - 20s : 9(3.6%), 30s : 75(30.0%), 40s : 107(42.8%), 50s and above : 59(23.6%))

Prior to conduct user expectation experience research, the

explanation for the definition of stretchable display and various types is provided, and time constraint was set so that the participants cannot start the survey and are asked to fully acquaint with the content. The survey questionnaires are composed of previously organized expectation experience items, and five-point Likert scale of ‘strongly agree’, ‘agree’, ‘neither agree nor disagree’, ‘disagree’, ‘strongly disagree’ is used to measure the level of expectation.

4.2 Confirmatory Factor Analysis and Result

Confirmatory factor analysis, one of statistical analysis methods of qualitative research, is used to analyze data. The confirmatory factor analysis is to test the relation between measurement variable (questionnaire: attributes) and latent variable (factor) and the relation between latent variables, and the analysis is done when the research model is set. The research model should be proposed based on the theoretical background or preceding research, and when the model is not appropriate, the variable is removed or Modification Index (MI) is used to select the final model. Confirmatory factor analysis has the characteristic of theory testing procedure by going through additional processes such as model fit test, Convergent validity and Discriminant validity, etc. to test the research model fits the data[37].

The confirmatory factor analysis on the initial model showed that standardized regression weight (S.R.W) is less than 0.5 , meaning it does not have convergent validity; thus, to improve the model, ‘F5_2: simple control process’ and ‘F5_3: tactual control utilization’ of ‘Factor 5(control convenience)’ and ‘F7_5: entertainment’ of ‘Factor 7(leisure activity)’ are eliminated because their S.R.W is less than 0.5. With the updated model, using AVE(average variance extracted) to test convergent validity, and it could be said to have convergent validity by verifying all factors have AVE greater than 0.5.

In addition, to improve model fidelity, the covariance relation was tested using modification indices, and the final model was fixed without ‘F4_2: new technology’ of ‘Factor4(novelty)’ and ‘F5_7: controlling behavior’ of ‘Factor 5(control convenience)’ and ‘F7_8: comfort of ‘Factor 7(relaxation)’ that either have covariability with other factors or have covariability with items within same factor. The confirmatory factor analysis on the final model is as follows(Table 4).

Table 4 Result of confirmatory factor analysis

F.no	A.no	S.R.W	S.E.	p-value	E.V.	AVE
F 1	F1_1	0.62	0.043	<0.001	0.233	0.578
	F1_2	0.615	0.049	<0.001	0.381	
	F1_3	0.621	0.049	<0.001	0.369	
	F1_4	0.648	0.043	<0.001	0.216	
	F1_5	0.616	0.046	<0.001	0.296	
	F1_6	0.682	0.046	<0.001	0.266	
F 2	F2_1	0.577	0.041	<0.001	0.225	0.586
	F2_2	0.623	0.047	<0.001	0.304	
	F2_3	0.563	0.044	<0.001	0.289	
	F2_4	0.63	0.041	<0.001	0.196	
F 3	F3_1	0.609	0.044	<0.001	0.262	0.533
	F3_2	0.509	0.044	<0.001	0.321	

	F3_3	0.572	0.043	<0.001	0.269	
	F3_4	0.574	0.044	<0.001	0.277	
F 4	F4_1	0.638	0.045	<0.001	0.231	0.634
	F4_3	0.64	0.045	<0.001	0.24	
F 5	F5_1	0.551	0.042	<0.001	0.268	0.563
	F5_4	0.581	0.042	<0.001	0.253	
	F5_5	0.564	0.041	<0.001	0.237	
	F5_6	0.568	0.043	<0.001	0.27	
	F5_8	0.59	0.042	<0.001	0.236	
F 6	F6_1	0.585	0.046	<0.001	0.312	0.511
	F6_2	0.553	0.045	<0.001	0.325	
	F6_3	0.568	0.045	<0.001	0.299	
	F6_4	0.593	0.048	<0.001	0.337	
	F6_5	0.596	0.046	<0.001	0.303	
	F6_6	0.534	0.044	<0.001	0.305	
F 7	F7_1	0.609	0.044	<0.001	0.268	0.579
	F7_2	0.669	0.045	<0.001	0.244	
	F7_3	0.52	0.048	<0.001	0.402	
	F7_4	0.606	0.045	<0.001	0.295	
	F7_6	0.669	0.045	<0.001	0.262	
	F7_7	0.697	0.047	<0.001	0.266	
F 8	F8_1	0.607	0.046	<0.001	0.302	0.566
	F8_2	0.584	0.044	<0.001	0.279	
	F8_3	0.681	0.05	<0.001	0.352	
	F8_4	0.642	0.047	<0.001	0.301	
	F8_5	0.613	0.044	<0.001	0.267	
F9	F9_1	0.579	0.05	<0.001	0.397	0.516
	F9_2	0.512	0.045	<0.001	0.325	
	F9_3	0.605	0.046	<0.001	0.297	
	F9_4	0.655	0.047	<0.001	0.285	
F 10	F10_1	0.577	0.042	<0.001	0.204	0.565
	F10_2	0.542	0.045	<0.001	0.278	
F11	F11_1	0.603	0.04	<0.001	0.181	0.645
	F11_2	0.599	0.044	<0.001	0.269	
	F11_3	0.651	0.041	<0.001	0.18	
F12	F12_1	0.677	0.043	<0.001	0.191	0.736
	F12_2	0.726	0.044	<0.001	0.162	
F13	F13_1	0.618	0.046	<0.001	0.302	0.535
	F13_2	0.625	0.045	<0.001	0.272	
	F13_3	0.565	0.045	<0.001	0.322	
	F13_4	0.525	0.041	<0.001	0.261	
	F13_5	0.536	0.043	<0.001	0.283	

$x^2(p) < .001$, $x^2: 2015.75$, RMSEA: .047, CFI: .918, TLI: .909

S.R.W: Standardized Regression Weight / S.E.: Standard Error / E.V. : Error Variance / AVE : Average Variance Extracted

Verification on final model is done using x^2 , which is an absolute fit index, Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI) that are incremental fit indices, and Root Mean Square Error of Approximation (RMSEA). The model fit index is used to verify the suitability of covariance matrix of the collected data(sample) and the estimated covariance matrix of the research model, the distribution of the covariance matrix of the sample and the research model is studied. The difference in x^2 to compare the models in nested relations was significant with the $x^2(p)$ value at the level of 0.0001. RMSEA value was .047; when the value less than .1, the fidelity is secured, and when less than .05, the model is

considered to have high fidelity. CIF was .918 and TLI was .909; with value greater than .90, the model is considered to have high fidelity[38].

All factors have AVE higher than .5, when calculated for each factor (latent variable), and this shows the convergent validity. It can be said that discriminant validity is also secured since the value obtained by adding and subtracting the correlation coefficient and 2 times the standard error of the covariance is not 1.

The reliability analysis is conducted to see the items consisting a factor have measurement consistency. The liability (Cronbach's alpha) of the items of each factor was Factor1: 0.890, Factor2: 0.848, Factor3: 0.817, Factor4: 0.776, Factor5: 0.865, Factor6: 0.861, Factor7: 0.889, Factor8: 0.867, Factor9: 0.808, Factor10: 0.721, Factor11: 0.843, Factor12: 0.847, Factor13: 0.850. It can be said that items have enough liability as liability threshold is greater than .7[39].

5 Discussion

The expectation experience factors derived from above research process is composed of 9 factors that are expanded from the existing automotive field and new 4 factors. The expectation experience factors and their meanings based on the finalized factors and attributes through the factor analysis and liability test are as following.

First factor is 'safety'. The attributes forming this factor are 'control safety(F1_1)', 'durability(F1_2)', 'Eye breakaway prevention(F1_3)', 'emergency preparation(F1_4)', 'anxiety relief(F1_5)', 'trust(F1_6)'. For safety, on HMI guideline of Automated Driving Systems: A Vision for Safety, US stated that the Automated Driving Systems not only need to perform safe driving feature but also clearly inform the intension of ADAS and vehicle performance information to passengers and pedestrians. Stretchable display is safe because it is unbreakable upon impact, can provide information on various spots with shape deformability, and especially has an advantage of being able to provide tactual information on top of visual information. With these characteristics, the stretchable display applied on vehicle is expected to be able to increase the trust and release anxiety by helping faster and more stable control and effective information acknowledgement

Second factor is 'high quality'. The attributes forming this factor are 'luxurious material(F2_1)', 'high end(F2_2)', 'quality of technique(F2_3)', 'superior(F1_4)'. The users expected the stretchable display to have superior material, as they understand it as a material for interface that can cover various curved surface of the vehicle. The stretchable display with the high technology is expected to be loaded to premium vehicles on its early stage, so it needs not only to have remarkable technology but also to deliver luxury from the visual and tactual quality property of the interface.

Third factor is about 'multi-sensory'. The attributes forming this factor are 'five sense stimulus experience(F3_1)', 'interaction using five senses(F3_2)', 'sensory(F3_3)', 'emotional(F3_4)'. The stretchable display's ability to

transform Z-axis enables the change in texture, and tactual expression of the interface can be used to deliver emotional experience. Visual/auditory and tactual sensations are provided to provide a multi-sensory experience. In the automotive field, sensual/emotional experience in addition to functional experience and customer value derived customer value from triggered emotion are shown to be important, so the stretchable display is expected to take a role to fulfill this expectation experience.

Forth factor is 'pleasure of novelty'. The attributes forming this factor are appeared to be 'pleasure of new way of control(F4_1)', 'novelty(F4_3)'. The stretchable display can do transformation interaction of stretching or twisting, and 3D interaction such as pushing in. This new interface gives freshness to the user as it is or fulfill their interest. Thus, the stretchable display designer needs to provide the enjoyment from the novelty by providing new user interface utilizing this new form property.

Fifth factor is 'contextual control convenience'. The attributes forming this factor are 'function integration within operational range(F5_1)', 'intuitive control(F5_4)', 'circumstance customization(contextual)(F5_5)', 'environment development (F5_6)', 'leading usability(F5_8)'. Recently the physical button of the control interface has been replaced with the soft button, and the soft button can be programmed to a suitable button based on functional usage context. On the other side of this advantage, it has a disadvantage that false control can occur because of the lack of perceptual information on the soft button. With the Z-axis height change and textual transformation property, the stretchable display is expected to overcome the limitations of the soft button on the flat display.

Sixth factor is 'physical/mental freedom'. The attributes forming this factor are 'relieve stress of driving(F6_1)', 'admire scenery(F6_2)', 'freedom(F6_3)', 'transport without having to care(F6_4)', 'perceived ease of use(F6_5)', 'emotion and energy management (F6_6)'. From the self-driving vehicle, the users expect to free time for viewing contents or admiring scenery away from physical/mental stress. When the driving is fully automated, the user does not need to look ahead, but in case the user concerns of the information perception and risk situation handling, it is had to be physically/mentally free. As the stretchable display can transform to various shapes, it can cover various spaces such as vehicle windshield, A pillar, seat cover, door, etc., necessary information and control interface can be promptly provided to the user in case of emergency. This means that the user does not have to standby or look ahead to prepare for emergency, and it is expected to provide physical/mental freedom to the user.

Seventh factor is 'relaxation service spatiality'. The attributes of this factor are 'relaxation/convenience space support(F7_1)', 'relaxation/convenience service support(F7_2)', 'spatiality improvement for storage(F7_3)', 'provide independent space(F7_4)', 'resting while travelling(F7_6)', 'relaxation(F7_7)'. It was found that the self-driving vehicle user wants to enjoy the relaxation and convenience service experience when the driving is fully automated and to be

provided with the service interface that can support these experiences. It is expected to be able to create sensible relaxation space through interior design changes, as interior display can cover various curved surfaces, as well as the service function side.

Eighth factor is 'productive activity spatiality'. The attributes of this factor are 'Business/meeting space support(F8_1)', 'amenities support(F8_2)', 'multitask (F8_3)', 'productive while travelling(F8_4)', 'time management and efficiency(F8_5)'. In the self-driving vehicle, the user expects to use time effectively by doing various activities supporting business task and life while travelling. The enlarged display, which covers wider space of the vehicle interior, can provide more various services. It is expected to improve the spatiality with the various contextual interior design as well as various business/life support service contents provided for the user's needs.

Ninth factor is 'uniqueness'. The attributes of this factor are 'self-expression(F9_1)', 'uniqueness (F9_2)', 'customized(F9_3)', 'suitability (F9_4)'. Unlike physical display, digital play can provide various function button or Look & Feel image for context. Since stretchable display can replace various physical interfaces, it became possible to provide interface suitable to the user using more diverse methods. The new-generation user's needs for expressing uniqueness and personalization are becoming more important (*), and as the artificial intelligence technology that provides user preference customization expands, the automotive interface hyper-personalization would gain speed. It is expected that there would be more diverse ways to customize vehicle functions or interior design or show uniqueness using the stretchable display.

Tenth factor is 'aesthetical fashion'. The attributes forming this factor are 'color and pattern variation(F10_1)', 'neat interior design (F10_2)'. It seems that the stretchable display is expected to be utilized as interior interface covering inner space of the vehicle. With the stretchable display's ability to express the color, patten and texture changes, the user expects to have aesthetically great experience from it. The physical interior design is difficult to make a change from the initial selections, but for soft interior design, various design needs to be provided so that the various variation can be made.

Eleventh factor is 'interface expandability'. The attributes forming this factor are 'multi-interface(F11_1)', 'digitalization of physical space(F11_2)', 'display area expansion(F11_3)'. Changing physical interface to digital interface is an expectation attribute which was emerged when previously irreplaceable various curved surface's interface was enabled by the stretchable display. Thus, the digital display area is expandable, and more multi functions are expected to be provided. With the stretchable display widely applied inside of vehicle, the digital interface area that can interact with the user is expected to be expanded.

Twelfth factor is 'continuous improvement'. The attributes forming this factor are 'software update(F12_1)', 'software utilization(F12_2)'. As the vehicles become electrified, in-vehicle functions could be updated using OTA (Over The

Air) method; as a result, the user could use continuously improved functions and the user interface on the digital display. The stretchable display expands the digital interface area that user can access and the user is expecting to receive the continuous improvement on vehicle's various functions.

Thirteenth factor is 'human/object communication'. The attributes of this factor are 'external communication(F13_1)', 'internal communication(F13_2)', 'passenger status recognition(F13_3)', 'information shareability(F13_3)', 'device pairing(F13_4)'. This is an experience expecting the support for connection between humans, human and object, and between objects. When the interior of the vehicle is covered with the stretchable display, the service and interface that support smooth communication between human/object inside and outside of the vehicle are needed to be provided.

6 Conclusion and Future Work

This study is conducted to find out the experience factors and the detailed attributes of the factors that user expects when the stretchable display is applied to automotive field. The exploratory study was conducted by collecting user expectation experience and defining factor structure based on literature study, and confirmatory study was conducted for the positive verification. The first to ninth factors are expanded from the expected experienced from existing automotive field, and it can be understood that it is hoped the existing expectation experience on the automotive field is effectively supported using the stretchable display. The detailed experience attributes are organized; this allows the issue which the existing interface environment could not resolve can be resolved by the interface which the stretchable display is applied. The eleventh to thirteenth factors are the new user experience that only can be expected from vehicle with stretchable display. As the stretchable display take role to expand the point of contract where new electronic technology and the user meet, it is expected to contribute provide the user experience in new ways.

The stretchable display can have flexible designs on various curved surfaces, and the design can be applied to various fields beyond the existing display application industry fields such as laptops, TV, smartphone, etc. For the display market growth, a new field, such as the vehicle display, needs to be developed. The expectation on the product, service, and environment with the stretchable display technology would be the growth engine for and would become a key factor determining the design and the performance of the new generation ICT products in the future. In this study, derived user Expectation experience factors and detailed attributes can be provided as goals for designing the automotive interface applying the stretchable display, or it can be provided as an evaluation standard after designing. The user experience factors can be utilized as an index for the entire user experience design process, and the developer can utilize it to set the direction of technology development. The study on the user experience factors or attributes has difficulty in that the direct experience cannot be achieved when the subject is the future technology. However, because the necessity of

commercialization technology that works in the market is rising, the user experience study that can be applied to the prototype design in the development stage is needed as basic research. For the next study, through the user scenario methodology that utilized derived user experience factors as indices, the product/service that materializes the interface inside the vehicle would be designed.

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References

- [1] Jo, J., Han, S., and Jeon, S., *The evolution of display form factor innovation*, LG Economic Research Institute, pp.1-37. <http://www.lgeri.com/report/view.do?idx=19580>, 2017. (Accessed: 03 May 2020)
- [2] Im, H., *2019 KISTEP Top 10 Promising Technologies*, KISTEP Inl, Vol.28, pp.38-39, 2019.
- [3] KEIT, *Development of 20% stretchable backplane and light emitting pixel for stretchable display*, KEIT PD Issue Peport, Vol 19-9, pp.1-15, <https://itech.keit.re.kr/index.do#03040100> 2019. (Accessed: 11 June 2021)
- [4] Lee, E., *The bigger the better.. Large-size automotive displays*, Munhwa Ilbo, <http://munhwa.com/mnews/view.html?no=2020021801032330322001>, 2020. (Accessed: 02 January 2021)
- [5] Ryu, J., *Samsung Electronics unveils 5G-based advanced driver's seat 'Digital Cockpit 2020'*, News1, <https://www.news1.kr/articles/?3811280>, 2020. (Accessed: 22 August 2020)
- [6] AEM, *Continental removes blind spots with A-pillar OLED*. Auto electronics Magazine. <http://www.autoelectronics.co.kr/article/articleView.asp?idx=3031>, 2019. (Accessed: 30 August 2020)
- [7] Netcarshow, *Mercedes-Benz F015 Luxury in Motion Concept 2015*, Net car show.com. https://www.netcarshow.com/mercedes-benz/2015-f015_luxury_in_motion_concept/, 2015. (Accessed: 30 August 2020)
- [8] Lee, W., *Why did Tesla drivers lose their eyes to the display and cause an accident?*. Motor Graph. https://www.netcarshow.com/mercedes-benz/2015-f015_luxury_in_motion_concept/, 2020. (Accessed: 22 August 2020)
- [9] Hong, J. H., Shin, J. M., Kim, G. M., Joo, H., Park, G. S., Hwang, I. B., Kim, M. W., Park, W. S., Chu, H. Y., and Kim, S., *9.1-inch stretchable AMOLED display based on LTPS technology*, Journal of the Society for Information Display, Vol.25, No.3, pp.194-199, 2017.
- [10] Kim, S. *"Freedom like a rubber band" Samsung Electronics develops 'stretchable display'...Expanded to freeform display*, Seoul Wire.

- <http://www.seoulwire.com/news/articleView.html?idxno=4446>, 46, 2021. (Accessed: 03 May 2021)
- [11] Jo, M., *LG Display to develop flexible and deformable free-form displays by 2024*. Work Today. <http://www.worktoday.co.kr/news/articleView.html?idxno=8117>, 2020. (Accessed: 11 June 2020).
- [12] Kim, D. H., Xiao, J., Song, J., Huang, Y., and Rogers, J. A., *Stretchable, curvilinear electronics based on inorganic materials*, *Advanced Materials*, Vol.22, No.19, pp.2108-2124, 2010.
- [13] Takahashi, T., Takei, K., Gillies, A. G., Fearing, R. S., and Javey, A., *Carbon nanotube active-matrix backplanes for conformal electronics and sensors*, *Nano letters*, Vol.11, No.12, pp.5408-5413, 2011.
- [14] Ahn, S. D., Kang, S. Y., Cho, S. H., Yang, J. H., Lee, H., Na, B. S., Koo, J. B., Park, C. W., Byun, C. W., Kwon, B. H., Cho, N. S., Hwang, C. S., and Lee, J. I., *Ultrathin Stretchable Oxide Thin Film Transistor and Active Matrix Organic Light-Emitting Diode Displays*, *SID Symposium Digest of Technical Papers*, Vol.48, No.1, pp.33-35, 2017.
- [15] Rogers, J. A., Someya, T., and Huang, Y., *Materials and mechanics for stretchable electronics*, *Science*, Vol.327, No.5973, pp.1603-1607, 2010.
- [16] Lee, S. S., Kim, S., Jin, B., Choi, E., Kim, B., Jia, X., Kim, D., and Lee, K. P., *How users manipulate deformable displays as input devices*, In *proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 1647-1656. 2010.
- [17] Müller, M., Knöfel, A., Gründer, T., Franke, I., and Groh, R., *Flexiwall: Exploring layered data with elastic displays*, In *proceedings of the Ninth ACM International Conference on Interactive Tabletops and Surfaces*, pp. 439-442, 2014.
- [18] Troiano, G. M., Pedersen, E. W., and Hornbæk, K., *User-defined gestures for elastic, deformable displays*, In *proceedings of the 2014 international working conference on advanced visual interfaces*, pp. 1-8, 2014.
- [19] Yun, K., Song, J., Youn, K., Cho, S., and Bang, H., *ElaScreen: exploring multi-dimensional data using elastic screen*, In *proceedings of CHI'13 Extended Abstracts on Human Factors in Computing Systems*, pp. 1311-1316, 2013.
- [20] Lee, S., Jeong, S., Park, C., Na, B., Oh, J., Koo, J., and Chu, H., *Stretchable Display Technology*, *Electronic communication trend analysis*, Vol.29, No.6, pp.63-70, 2014.
- [21] Holman, D., and Vertegaal, R., *Organic user interfaces: designing computers in any way, shape, or form*, *Communications of the ACM*, Vol.51, No.6, pp.48-55, 2008.
- [22] Campbel, M., *Apple Invents Stretchable Display Backed by Mesh-like Elastomer*, *Apple Insider*. <https://appleinsider.com/articles/17/02/09/apple-invents-stretchable-display-backed-by-mesh-like-elastomer>, 2017 (Accessed: 05 May 2020)
- [23] Shende, V., *Analysis of research in consumer behavior of automobile passenger car customer*, *International Journal of Scientific and Research Publications*, No.4, Vol.2, pp.1-8. 2014.
- [24] Tynan, C., McKechnie, S., and Hartley, S., *Interpreting value in the customer service experience using customer-dominant logic*, *Journal of marketing management*, Vol.30. No.9-10, pp.1058-1081, 2014.
- [25] Sari, D., and Kusuma, B., *Does luxury brand perception matter in purchase intention? A comparison between a Japanese brand and a German brand*. *ASEAN Marketing Journal*, Vol.6, No.1, pp.50-63, 2014.
- [26] Yi, S., *The era of machines driving is coming... What does autonomous driving level 3 mean?*, *KBS News*. <https://news.kbs.co.kr/news/view.do?ncd=5049953>, 2020. (Accessed: 05 May 2021)
- [27] Ministry of Land, Infrastructure and Transport, *Establishment and confirmation of the 1st autonomous driving transportation and logistics master plan ('21 - '25)*, http://www.molit.go.kr/USR/NEWS/m_71/dtl.jsp?id=95085718, 2021.
- [28] Jung, S. H., *Automobile electronic products, the fastest growing among electronic products*, *Yonhapnews*, <https://www.yna.co.kr/view/AKR20171111035600003>, 2017. (Accessed: 22 July 2020)
- [29] König, M., and Neumayr, L., *Users' resistance towards radical innovations: The case of the self-driving car*, *Transportation research part F: traffic psychology and behaviour*, Vol.44, pp.42-52, 2017.
- [30] Zmud, J. P., and Sener, I. N., *Towards an understanding of the travel behavior impact of autonomous vehicles*, *Transportation research procedia*, Vol.25, pp.2500-2519, 2017.
- [31] Howard, D., and Dai, D., *Public perceptions of self-driving cars: The case of Berkeley, California*. In *Proceedings of Transportation research board 93rd annual meeting*. Vol. 14, No. 4502, pp. 1-16, 2014.
- [32] Pettersson, I., *Travelling from fascination to new meanings: Understanding user expectations through a case study of autonomous cars*, *International Journal of Design*, Vol.11, No.2, pp. 1-11, 2017.
- [33] Rödel, C., Stadler, S., Meschtscherjakov, A., and Tscheligi, M., *Towards autonomous cars: The effect of autonomy levels on acceptance and user experience*, In *proceedings of the 6th international conference on automotive user interfaces and interactive vehicular applications*, pp. 1-8. 2014.
- [34] Jo, S. B., *This is UX/UI design*. Seoul, Korea: Wikibooks. 2020.
- [35] Strauss, A. and Corbin, J., *Basics of Qualitative Research* (2nd ed.). SAGE Publications, Inc. Seoul : Hyeonmunsa, 1998.
- [36] Hassenzahl, M., *The Thing and I: Understanding the Relationship Between User and Product*. In M.A. Blythe, K. Overbeeke, A.F. Monk, and P.C. Wright, (Eds). *Funology: From Usability To Enjoyment*, pp.31-42. Kluwer Academic Publishers, 2003.
- [37] Van Prooijen, J. W., and Van Der Kloot, W. A., *Confirmatory analysis of exploratively obtained factor structures*, *Educational and Psychological Measurement*, Vol.61, No.5, pp.777-792, 2001.
- [38] Choi, C. H., and You, Y. Y., *The study on the comparative analysis of EFA and CFA*. *Journal of Digital Convergence*,

Vol.15, No.10, pp.103-111, 2017.

[39] Kim, K., *Social Welfare Survey Methodology Lecture*,
Seoul, Korea: MJ Media, 2008.